

WALKTHROUGHS T2 -OPP CrealtivaStudio

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1. We don't have an option to go back in my menu

The application suffered from a UI/Usability flaw where the "REPORTES E INFORMES" (Reports and Information) sub-menu lacked an explicit option to return to the MENU PRINCIPAL (Main Menu). Although not a code crash, this trapped the user in the sub-menu, hindering navigation.

```

:: Menu. de factura #1 (000000) ver

--- MENU PRINCIPAL ---
1. Gestion de Clientes
2. Gestion de Eventos
3. Gestion de Facturas
4. Reportes e Informes
Seleccione una opcion: 4

--- REPORTES E INFORMES ---
1. Facturas pendientes de pago
2. Facturas pagadas
3. Resumen financiero
4. Volver al menu principal
Seleccione una opcion: |
```

Solution

The solution involved implementing necessary **navigation validations** to allow the user to return or exit.

- **Sub-Menu Fix:** The REPORTES E INFORMES sub-menu was updated to include **4. Volver al menu principal**.
- **Main Menu Enhancement:** The MENU PRINCIPAL was updated to explicitly include **5. Salir** (Exit) as the primary way to terminate the application gracefully.

```

--- MENU PRINCIPAL ---
1. Gestion de Clientes
2. Gestion de Eventos
3. Gestion de Facturas
4. Reportes e Informes
5. Salir
Seleccione una opción: 5
Gracias por usar Creative Studio Management System!
BUILD SUCCESSFUL (total time: 3 seconds)
|

```

2. Customer Registration Error

An error was encountered when registering a new customer on the line where the Customer object was created (Customer customer = new Customer(name, phone, email, address);).

```

1. Registrar nuevo cliente
2. Listar todos los clientes
3. Buscar cliente por ID
4. Buscar cliente por nombre
5. Actualizar cliente
6. Eliminar cliente
7. Ver detalles completos de cliente
8. Volver al menu principal
Seleccione una opción: 1
Exception in thread "main" java.lang.RuntimeException: Uncompilable code - illegal start of expression
    at creativastudio.view.CreativeStudioApp.registerCustomer(CreativeStudioApp.java:11)
    at creativastudio.view.CreativeStudioApp.manageCustomers(CreativeStudioApp.java:81)
    at creativastudio.view.CreativeStudioApp.main(CreativeStudioApp.java:42)
C:\Users\USER\AppData\Local\NetBeans\Cache\27\executor-snippets\run.xml:111: The following error occurred w
C:\Users\USER\AppData\Local\NetBeans\Cache\27\executor-snippets\run.xml:68: Java returned: 1
BUILD FAILED (total time: 12 seconds)

```

Solution

The issue was resolved by adding a **conditional validation (if/else)** to check the success of the data saving operation.

- **Saving Validation:** The logic was wrapped to execute the save command and check its boolean result: if (customer.save()).
- **Success Flow:** If customer.save() returns true, a success message is printed, along with the assigned customer ID (ID asignado:).
- **Error Flow:** If the save operation fails (returns false), an error message is displayed to the user: Error al registrar cliente. Verifique los datos. (Error registering customer. Verify data.).

```

        Customer customer = new Customer(name, phone, email, address,);
        if (customer.save()) {
            System.out.println(x: "Cliente registrado exitosamente!");
            System.out.println("ID asignado: " + customer.getId());
        } else {
            System.out.println(x: "Error al registrar cliente. Verifique los datos.");
        }
    }
}

```

3. Customer List Not Displaying

The option to 2. Listar todos los clientes (List all customers) failed to display the registered customers, producing the same Runtime Error: Uncompilable code illegal start of expression. This suggests the program was attempting to process or iterate over the customer list without correctly handling the scenario where the list might be empty or improperly accessed.

```

2. Listar todos los clientes
3. Buscar cliente por ID
4. Buscar cliente por nombre
5. Actualizar cliente
6. Eliminar cliente
7. Ver detalles completos de cliente
8. Volver al menu principal
Seleccione una opcion: 2
] Exception in thread "main" java.lang.RuntimeException: Uncompilable code - illegal start of expression
|   at creativastudio.view.CreativeStudioApp.listAllCustomers(CreativeStudioApp.java:1)
|   at creativastudio.view.CreativeStudioApp.manageCustomers(CreativeStudioApp.java:82)
|   at creativastudio.view.CreativeStudioApp.main(CreativeStudioApp.java:42)
C:\Users\USER\AppData\Local\NetBeans\Cache\27\executor-snippets\run.xml:111: The following error occurred while executing this line:
C:\Users\USER\AppData\Local\NetBeans\Cache\27\executor-snippets\run.xml:66: Java returned: 1
BUILD FAILED (total time: 6 seconds)

```

Solution

The solution implemented an **if and else validation** to safely manage the list retrieval and iteration.

- **List Retrieval:** The method first retrieves the list of customers: `List<Customer> customers = Customer.getAllCustomers();`.
- **Empty Check:** It checks if the list is empty: `if (customers.isEmpty())`. If true, it prints "No hay clientes registrados." (No registered customers) .
- **Safe Iteration:** If the list is not empty, it proceeds to safely iterate through the list using a for loop and prints the simple string representation of each customer , concluding with the total count.

```

private static void listAllCustomers() {
    System.out.println(x: "\n--- LISTA DE CLIENTES ---");
    List<Customer> customers = Customer.getAllCustomers();

    if (customers.isEmpty()) {
        System.out.println(x: "No hay clientes registrados.");
    } else {
        for (Customer customer : customers) {
            System.out.println(x: customer.toSimpleString());
        }
        System.out.println("\nTotal de clientes: " + customers.size());
    }
}

```

4. Non-Existent Customer ID Validation

When attempting an event operation, such as **2. Ver eventos de un cliente** (View customer events), the program crashed if the entered customerId did not exist. This caused a **Runtime Error: Uncompilable code illegal start of expression** because the code did not handle the null value returned by `Customer.findCustomerById(customerId)`.

```
-----  
--- GESTION DE EVENTOS ---  
1. Agregar evento a cliente  
2. Ver eventos de un cliente  
3. Eliminar evento  
4. Volver al menu principal  
Seleccione una opcion: 2  
Exception in thread "main" java.lang.RuntimeException: Uncompilable code - illegal start of expression  
at creativastudio.view.CreativeStudioApp.viewCustomerEvents(CreativeStudioApp.java:1)  
at creativastudio.view.CreativeStudioApp.manageEvents(CreativeStudioApp.java:216)  
at creativastudio.view.CreativeStudioApp.main(CreativeStudioApp.java:43)  
C:\Users\USER\AppData\Local\NetBeans\Cache\27\executor-snippets\run.xml:111: The following error occurred while executing this line:
```

Solution

A crucial explicit if (customer != null) validation was added immediately after attempting to find the customer.

- Null Check: The code now verifies if the customer object is successfully retrieved (not null).
- Logic Execution: If the customer is found (!= null), the intended function is executed (`customer.displayCustomerInfo()`).
- Error Messaging: If the customer is not found (is null), a friendly error message is displayed: "Cliente no encontrado." (Customer not found.) .

```
private static void viewCustomerEvents() {  
    int customerId = getIntInput(message: "Ingrese el ID del cliente: ");  
    Customer customer = Customer.findCustomerById(customerId);  
  
    if (customer != null) {  
        customer.displayCustomerInfo();  
    } else {  
        System.out.println(x: "Cliente no encontrado.");  
    }  
}
```

5. Input Type Conversion Failure (NumberFormatException)

When the user entered **non-numeric data** (like "asd" or "10") instead of a number when prompted to select a menu option, the application crashed with a **java.lang.NumberFormatException: For input string: "..."**. This exception occurs because the method **Integer.parseInt()** cannot convert text characters into an integer.

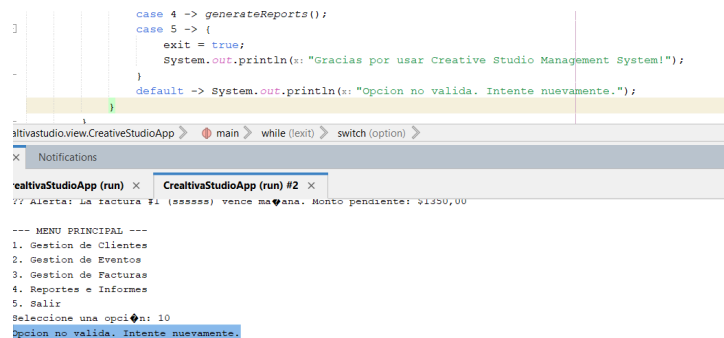
```
--- MENU PRINCIPAL ---
1. Gestion de Clientes
2. Gestion de Eventos
3. Gestion de Facturas
4. Reportes e Informes
5. Salir
Seleccione una opcion: 10

--- MENU PRINCIPAL ---
1. Gestion de Clientes
2. Gestion de Eventos
3. Gestion de Facturas
4. Reportes e Informes
5. Salir
Seleccione una opcion:
```

Solution

The solution involved implementing a try-catch block within the input method, `getIntInput(String message)`, to robustly handle invalid non-numeric input.

- **Error Handling:** The code that attempts to parse the integer (`Integer.parseInt(scanner.nextLine().trim());`) is placed inside a try block.
- **Input Loop:** A while (true) loop is used to continually prompt the user for input until a valid number is provided.
- **Exception Catching:** If a `NumberFormatException` occurs, the catch block executes, printing an error message: "Error: Por favor ingrese un numero valido." (Error: Please enter a valid number.) . The loop then restarts, asking the user to try again.



```
case 4 -> generateReports();
case 5 -> {
    exit = true;
    System.out.println("Gracias por usar Creative Studio Management System!");
}
default -> System.out.println("Opcion no valida. Intente nuevamente.");

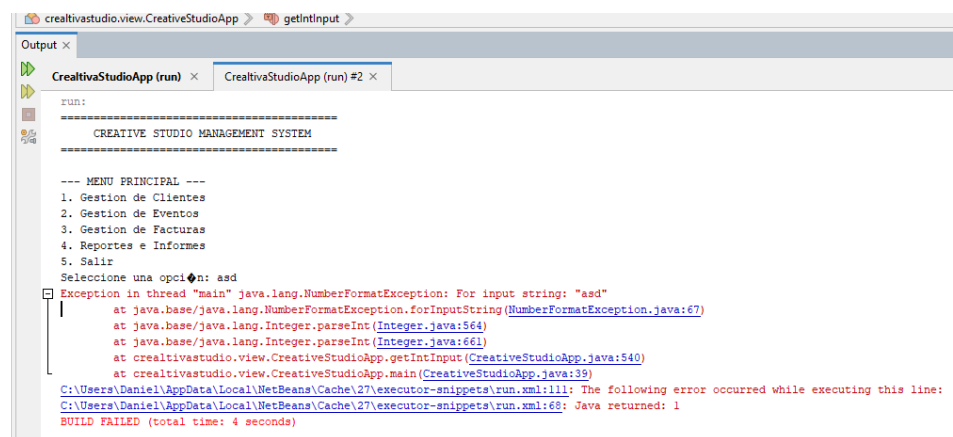
altivastudio.view.CreativeStudioApp > main > while (lexit) > switch (option) >
X Notifications
CreativeStudioApp (run) X CreativeStudioApp (run) #2 X
?? Alerta: La factura #1 (ssssss) vence mañana. Monto pendiente: $1350,00

--- MENU PRINCIPAL ---
1. Gestion de Clientes
2. Gestion de Eventos
3. Gestion de Facturas
4. Reportes e Informes
5. Salir
Seleccione una opcion: 10
Opcion no valida. Intente nuevamente.
```

6. Error Input Type Conversion Failure

This is a Runtime Error that occurs when the user enters non-numeric data (like "asd") instead of an integer when prompted to select a menu option.

- Exception Name: java.lang.NumberFormatException.
- Root Cause: The method getIntInput() attempts to use Integer.parseInt() to convert the user's string input (e.g., "asd") into an integer, but fails because the text is not a valid numeric format.
- Consequence: The program terminates and displays the stack trace.



```
creativastudio.view.CreativeStudioApp > getIntInput >
Output x
CreativeStudioApp (run) x CreativeStudioApp (run) #2 x
run:
=====
CREATIVE STUDIO MANAGEMENT SYSTEM
=====

--- MENU PRINCIPAL ---
1. Gestion de Clientes
2. Gestion de Eventos
3. Gestion de Facturas
4. Reportes e Informes
5. Salir
Seleccione una opción: asd
Exception in thread "main" java.lang.NumberFormatException: For input string: "asd"
    at java.base/java.lang.NumberFormatException.forInputString(NumberFormatException.java:67)
    at java.base/java.lang.Integer.parseInt(Integer.java:564)
    at java.base/java.lang.Integer.parseInt(Integer.java:661)
    at creativastudio.view.CreativeStudioApp.getIntInput(CreativeStudioApp.java:540)
    at creativastudio.view.CreativeStudioApp.main(CreativeStudioApp.java:39)
C:\Users\Daniel\AppData\Local\NetBeans\Cache\27\executor-snippets\run.xml:111: The following error occurred while executing this line:
C:\Users\Daniel\AppData\Local\NetBeans\Cache\27\executor-snippets\run.xml:68: Java returned: 1
BUILD FAILED (total time: 4 seconds)
```

Solution

The solution involves implementing a try-catch block within the getIntInput(String message) method to handle invalid input robustly .

- Error Handling: The parsing logic (Integer.parseInt()) is placed inside a try block .
- Exception Catching: If a NumberFormatException occurs, the catch block executes, printing an error message: "Error: Por favor ingrese un numero valido." (Error: Please enter a valid number.) .
- Input Loop: The method remains inside a while (true) loop, ensuring the user is prompted repeatedly until a valid number is entered.

```
private static int getIntInput(String message) {
    while (true) {
        try {
            System.out.print(message);
            return Integer.parseInt(scanner.nextLine().trim());
        } catch (NumberFormatException e) {
            System.out.println("Error: Por favor ingrese un numero valido.");
        }
    }
}
```

7. Failure to Access Null Object Property.

This is a Runtime Error that occurs when the program attempts to perform an operation on an object variable that currently holds the value null.

- Exception Name: java.lang.NullPointerException.
- Root Cause: The error happens when the user attempts to create a bill and enters a customerId that does not exist (e.g., ID 32). The search method (Customer.findCustomerById()) returns null.
- Specific Message: "Cannot invoke... Customer.getEvents() because customer is null". The code crashes attempting to access the list of events (getEvents()) on a non-existent customer object.
- Location of Crash: Inside the createBill method

```
8. Ver detalles de factura
9. Volver al menu principal
Seleccione una opcion: 1

--- CREAR NUEVA FACTURA ---
ID del cliente: 32
Exception in thread "main" java.lang.NullPointerException: Cannot invoke "creativastudio.model.Customer.getEvents()" because "customer" is null
    at creativastudio.view.CreativeStudioApp.createBill(CreativeStudioApp.java:332)
    at creativastudio.view.CreativeStudioApp.manageBills(CreativeStudioApp.java:305)
    at creativastudio.view.CreativeStudioApp.main(CreativeStudioApp.java:44)
C:\Users\Daniel\AppData\Local\NetBeans\Cache\27\executor-snippets\run.xml:111: The following error occurred while executing this line:
C:\Users\Daniel\AppData\Local\NetBeans\Cache\27\executor-snippets\run.xml:68: Java returned: 1
BUILD FAILED (total time: 19 seconds)
```

Solution

The solution is to add an explicit if (customer == null) validation immediately after retrieving the customer object in the createBill() method.

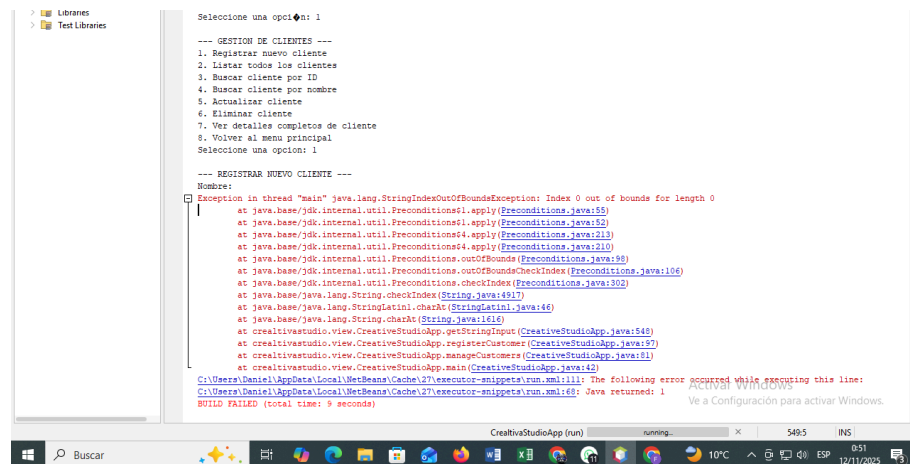
- Validation: The code checks if the customer object is null.
- Controlled Flow: If customer is null, the program prints "Cliente no encontrado." (Customer not found.) and exits the function gracefully using return. This prevents the execution from reaching the line that attempts to call customer.getEvents(), thus avoiding the NPE.

```
if (customer == null) {
    System.out.println("Cliente no encontrado.");
    return;
}
```

8. Index Failure While Attempting to Read Character from Empty String.

This is a Runtime Error that occurs when the code tries to access a character at an index that is outside the valid range of a string.

- Exception Name: `java.lang.StringIndexOutOfBoundsException`.
- Root Cause: The error occurs when the user is prompted for string input (e.g., Name when registering a new customer) and presses Enter without typing anything.
- Specific Message: "Index 0 out of bounds for length 0". This indicates the code was attempting to access the first character (`charAt(0)`) of a string that had a length of zero.



Solution

The solution is to ensure the `getStringInput()` method does not attempt any index access operation that would fail on an empty string. The corrected version of the method is the simple and safe implementation:

- **Clean Implementation:** The method is reset to simply prompt the user and return the trimmed line of input:

```
private static String getStringInput(String message) {  
    System.out.print(message);  
    return scanner.nextLine().trim();  
}
```